Spring 2025: CSE 434 **Programming Languages** **Akdeniz University**

16/04/2025 Midterm Exam Duration: 90 minutes

Name: _____ Student No: _____

P1 [30 points] Multiple Choice (Circle the correct answer) (Correct +2, Empty 0, Wrong -0.5) (Note: Negative scores apply only to this section. Your score here cannot drop below 0.)

1. Which of the following is a key criterion for language readability?
(a) Dynamic typing (b) Syntax consistency (c) Compile time (d) Symbol overloading

2. What language was the first high-level programming language to be implemented?
(a) ALGOL 60 (b) Lisp (c) COBOL (d) Fortran I

3. Which of the following describes a language with referential transparency?
(a) Its variables can be reassigned freely (b) Functions yield the same output for the same input (c) It supports pointers (d) It allows side-effects in all expressions

4. Which language was originally designed to support AI applications using symbolic computation?
(a) Prolog (b) Scheme (c) Java (d) Lisp

5. Which of the following is a weakness of ALGOL 60?
(a) Lack of recursion (b) Poor array support (c) No I/O capabilities (d) Lack of floating-point arithmetic

6. The evaluation of `(CAR (CDR '(A B C)))` in Scheme returns:
(a) A (b) B (c) (B C) (d) C

7. The main advantage of using BNF to describe syntax is:
(a) Faster execution (b) Easier I/O (c) Clear grammar definition (d) Object orientation

8. A grammar that allows more than one parse tree for the same sentence is called:
(a) Binary (b) Forked (c) Ambiguous (d) BNF

9. Which of the following constructs is typical of functional programming languages?
(a) Iterative loops (b) Assignment statements (c) Goto statements (d) First-class functions

10. In Scheme, what does `(DEFINE (square x) (* x x))` do?
(a) Assigns a value to `square` (b) Binds a value to `x` (c) Defines a function (d) Creates a variable `square`

11. What does an operational semantics approach describe?
(a) Program structure (b) Execution on a machine (c) Lexical rules (d) Type inference

12. Which language was the first to support data abstraction through classes?
(a) Java (b) C++ (c) Smalltalk (d) SIMULA 67

13. What is the main purpose of a lexical analyzer?
(a) Build the parse tree (b) Tokenize source code (c) Detect type errors (d) Manage memory

14. What is the purpose of the `QUOTE` function in Scheme?
(a) Converts atoms to strings (b) Prevents evaluation (c) Displays an expression (d) Compares two lists

15. Which language contributed the first pointer data type and exception handling?
(a) C (b) Java (c) PL/I (d) Ada

1.	A	B	C	D
2.	A	B	C	D
3.	A	B	C	D
4.	A	B	C	D
5.	A	B	C	D
6.	A	B	C	D
7.	A	B	C	D
8.	A	B	C	D
9.	A	B	C	D
10.	A	B	C	D
11.	A	B	C	D
12.	A	B	C	D
13.	A	B	C	D
14.	A	B	C	D
15.	A	B	C	D

P2 [10 points] Compute the weakest precondition for each of the following sequences of assignment statements and their postconditions:

```
(a) x = y + 3;
    y = x / 2
    {y > 5}
```

```
(b) a = b + 2 * a;
    b = 3 * a
    {b < 30}
```

P3 [15 points] Given the following scheme code:

```
(define (fr n d) (/ (- n (modulo n d)) d))
(define (td n) (modulo (fr n 10) 10))
```

a) (td 348) evaluates to _____.

b) (map td '(5 23 161 98 1070)) evaluates to _____.

c) (filter (lambda (x) (= (td x) 10)) '(14 24 37 43 45 55 60)) evaluates to _____.

P5 [10 points] Consider the following grammar and find out which of the following strings are in the language generated by this grammar? (Circle correct options.)

⟨S⟩ → a⟨S⟩b | a⟨S⟩bb | ⟨S⟩⟨S⟩ | ε

abbb Yes No ababab Yes No abaabbb Yes No aaabbbbbb Yes No

P6 [20 points] Consider the grammar and the corresponding LR parsing table from the textbook. Write the trace of a parse of the string id+(id+id)

1. E → E + T
2. E → T
3. T → (E)
4. T → id

State	Action					GoTo	
	id	+	(	)	$	E	T
0	S4		S3			1	2
1		S5			accept		
2	R2	R2	R2	R2	R2		
3	S4		S3			6	2
4	R4	R4	R4	R4	R4		
5	S4		S3				8
6		S5		S7			
7	R3	R3	R3	R3	R3		
8	R1	R1	R1	R1	R1		

Stack	Input	Action
0	id+(id+id)$	S4